# WILLIAM O. HOLDEN

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## SUMMARY

AI Engineer building internal agentic and retrieval-augmented applications at Aderant for the teams behind the company's legal software. Background spans full-stack development, LLM orchestration, and deep learning for computer vision and NLP, with peer-reviewed research in *Computers and Electronics in Agriculture*.

## PROFESSIONAL EXPERIENCE

**Aderant** Dec 2025 – Present
*AI Engineer, Docketing & VI Engineering* *Atlanta, GA (Hybrid)*

- Build internal AI applications for the teams behind Aderant's legal software, focused on agentic workflows and retrieval-augmented generation.
- Stack: Python, TypeScript/React, AWS, Claude Platform, Terraform.

**University of Georgia, School of Computing** 2024 – Dec 2025
*Graduate Research Assistant, Computer Vision* *Athens, GA*

- Built a spatiotemporal video encoder with metadata fusion (X3D 3D-CNN + YOLO tracking) for automated broiler-chicken behavior recognition from smartphone farm footage: 86% test accuracy (macro F1 0.86) across five behaviors.
- Developed motion-correction algorithms to normalize avian kinematics and remove human-handling bias, enabling reliable velocity and acceleration metrics from handheld video.
- 1st Place, UGA Artificial Intelligence in Livestock and Poultry Farming Conference poster competition (Sep 2025).

## SELECTED PROJECTS

**DeckTection: Trading Card Identification Pipeline** PyTorch, YOLOv12, SAM2

- Engineered an end-to-end CV pipeline (YOLOv12 → SAM2 → Siamese network); fine-tuned YOLOv12 to 99.49% mAP50 and trained a ResNet-18 Siamese network to 95% one-shot accuracy across 4,000 cards.

**Whisper Inference Optimization** ONNX Runtime, Quantization

- Quantized the Whisper encoder to INT8, cutting model size 70% with no material word-accuracy loss (LibriSpeech); implemented Winograd convolution for 1D layers for 60% compression enabling mobile deployment.

**Evaluating LLMs as Translators** MATEO, COMET, BLEU

- Benchmarked 15 Korean→English systems (DeepL, Google, GPT-5, Claude, DeepSeek, local Llama/Qwen/Gemma models) against the published human translation; DeepL led COMET (86.5) while Claude, DeepSeek, and GPT-5 beat it on BLEU and chrF2.

**Cinema E-Booking Platform** React, Django, SQL

- Full-stack booking system; led Agile sprints for a 5-engineer team.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript/TypeScript, Java, C++, C, SQL, R
**AI & ML:** Agentic workflows & LLM orchestration, RAG, vector search, prompt engineering, Claude Platform, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, SAM, ONNX
**Cloud & DevOps:** AWS (EC2, ECS, EKS, Lambda, S3, Route 53, Aurora/RDS), Terraform, Flux GitOps, Docker, Grafana, Git/GitHub Actions
**Data & Web:** React, Node.js, Django, REST APIs, Aurora/PostgreSQL, MongoDB, SQL Server
**Tools:** Glean, Browserbase, Playwright, Postman

## EDUCATION

**University of Georgia**, M.S. in Artificial Intelligence, GPA 3.73 Dec 2025
**University of Georgia**, B.S. in Computer Science, Honors Program, Presidential Scholar, Dean's List Dec 2024

## PUBLICATIONS

Holden, W.O., Li, G., Agyemang-Duah, E., et al. "Spatiotemporal video encoder integrated with metadata fusion for action recognition and behavior analysis." *Computers and Electronics in Agriculture* 253 (2026), 112159. doi:10.1016/j.compag.2026.112159 (IF 10.3).

Agyemang-Duah, E., Poku, R.A., Adenyo, C., Osei-Amponsah, R., Holden, W.O., et al. "Climate adaptation in poultry: Spatiotemporal behavioral and kinematic responses of chicken genotypes to identical temperature-humidity exposures." *Journal of Veterinary Behavior* 87 (2026), 41–49. doi:10.1016/j.jveb.2026.08.002.